

PHY 456 Tur 6 :

## Spin- $\frac{1}{2}$ Evolution

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Today : Pauli Hamiltonian, evolution of spin- $\frac{1}{2}$ .

Derivation: see nonrelativistic limit of Dirac Hamiltonian.

Motivation: EM fields (classical) are described by a scalar and vector potential

$$\phi(\vec{x}, t)$$

$$\vec{A}(\vec{x}, t).$$

$$\vec{E} = -\nabla\phi - \frac{\partial\vec{A}}{\partial t} \quad \vec{B} = \nabla \times \vec{A}.$$

What would like to ask

(i) What is the Hamiltonian operator for a

spinor wavefunction in an electromagnetic field

(ii) How does relate to the classical Lorentz

force law  $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$  Ehrenfest's Theorem.

Recall that a spinor can be written as a 2-component wavefunction  $\psi(\vec{x}) = \begin{pmatrix} \psi_+(\vec{x}) \\ \psi_-(\vec{x}) \end{pmatrix}$  correspond to  $S_z$  spin up/down

$$S_z \psi_+ = m_s \hbar \psi_+ \quad m_s = +\frac{1}{2}$$

$$S_z \psi_- = m_s \hbar \psi_- \quad m_s = -\frac{1}{2}.$$

Hence  $\psi_{\pm}(\vec{x}) = \langle \vec{x} | \pm \rangle$  position representation of  $|\pm\rangle_z$  basis.

$$\text{or } \psi(\vec{x}) = \psi_+(\vec{x}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \psi_-(\vec{x}) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hilbert Space  
spanned by  $|\pm\rangle$   
is isomorphic to  $\mathbb{C}^2$

SE general form:  $i\hbar \frac{\partial \psi(\vec{x}, t)}{\partial t} = \hat{H} \psi(\vec{x}, t).$

$$\begin{aligned} \text{Recall } \hat{H} &= \frac{\hat{p}^2}{2m} + V(\vec{x}) \\ &= -\frac{\hbar^2}{2m} \nabla^2 + V(\vec{x}, t) \end{aligned}$$

The magnetic field cannot be written as the gradient

of a potential, so we must modify our momentum operator (of some  $V(\vec{x}, t)$ ).

In electrostatics, we can always write  $\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$  = 0 if static

This implies that the potential associated with electrostatics

$$\Rightarrow q\phi(\vec{x}) = V(\vec{x}). \quad \vec{F} = -\nabla V \Rightarrow \vec{F} = q\vec{E}.$$

Static field:  $\hat{H} = \frac{\hat{p}^2}{2m} + q\phi(\vec{x})$

What about a magnetic field? See Dirac eqn.

$$\hat{p} \mapsto \hat{p} - q\vec{A}$$

which is called **minimal coupling**: the minimum requirement to incorporate a magnetic field in the Hamiltonian

$$\vec{\hat{p}} \rightarrow \vec{\hat{p}} - q\vec{A}$$

$$\partial_\mu \rightarrow \partial_\mu - \frac{iq}{\hbar} A_\mu$$

$A_\mu \equiv \text{field}$

Thus  $\hat{H} = \frac{1}{2m} (\vec{\hat{p}} - q\vec{A})^2 + q\phi$

Spin 0 coupled to  $\vec{E}/\vec{B}$  field

$(\vec{x}, t)$  dependent now that we've included  $\vec{A}$ .

$$\hat{H} = -\frac{1}{2m} (\hat{p} - q\vec{A})^2 + q\phi.$$

How do we get spinors out of this?

To add spin  $\frac{1}{2}$  [ $S = \frac{1}{2}$ ]:

For an electron fixed in space, in a constant  $\vec{B}$ -field, the Schrödinger equation is

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{q}{m} \hat{S} \cdot \vec{B} \Psi(\vec{x}, t) \dots$$

$$= \underbrace{-\frac{q\hbar}{2m}}_{\text{Magnetic Moment}} \hat{\sigma} \cdot \vec{B} \Psi(\vec{x}, t)$$

↓ Magnetic Moment

$-\vec{\mu} \cdot \vec{B}$  which is the classical Hamiltonian.

But what about the Hamiltonian?

We might try

$$\sigma_0 = \mathbb{I}_{2 \times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{H} = \frac{1}{2m} (\hat{p} - q\vec{A})^2 \cdot \sigma_0 + q\phi \sigma_0 - \frac{q\hbar}{2m} \hat{\sigma} \cdot \vec{B}$$

spin "mixing" term

"Remarkably, this is correct!"

where  $\vec{B} = \vec{B}(\vec{x}, t)$  now.

Upon taking the nonrelativistic limit of Dirac-Maxwell equation (Dirac can be coupled to Maxwell minimally, not QED) one finds

$$\hat{H} = \frac{1}{2m} [\vec{\sigma} \cdot (\hat{\vec{p}} - q\vec{A})]^2 + q\phi(\vec{x}, t)$$

Pauli  
HAMILTONIAN

From HW2:

$$(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = (\vec{A} \cdot \vec{B}) \sigma_0 + i\vec{\sigma} \cdot (\vec{A} \times \vec{B})$$

$$\vec{A} = \vec{B} = \hat{\vec{p}} - q\vec{A} = -i\hbar\nabla - q\vec{A}$$

$$[\vec{\sigma} \cdot (\hat{\vec{p}} - q\vec{A})]^2 = (\hat{\vec{p}} - q\vec{A})^2 \sigma_0 + i\vec{\sigma} \cdot (-i\hbar\nabla - q\vec{A}) \times (-i\hbar\nabla - q\vec{A})$$

$\uparrow$  here this                       $\uparrow$  what's this?

only nonzero terms are  
cross terms

For a general  $\psi$ , we have

$$\begin{aligned}
 (-i\hbar\nabla - q\vec{A}) \times (-i\hbar\nabla - q\vec{A}) \psi &\stackrel{\text{curl identity}}{=} -q [\hat{\vec{p}} \times (\vec{A}\psi) + \vec{A} \times (\hat{\vec{p}}\psi)] \\
 &= i\hbar q [\nabla \times (\vec{A}\psi) + \vec{A} \times (\nabla\psi)] \\
 &= i\hbar q [(\nabla \times \vec{A})\psi - \vec{A} \times (\nabla\psi) + \cancel{\vec{A} \times \nabla\psi}]
 \end{aligned}$$

$$= i\hbar q \vec{B} \cdot \vec{\sigma}$$

$\times i\omega$

We find

$$[\vec{\sigma} \cdot (\vec{p} - q\vec{A})]^2 = (\vec{p} - q\vec{A})^2 \sigma_0 - i\hbar q \vec{\sigma} \cdot \vec{B}$$

but multiplied by  $\frac{1}{2m}$  gives

$$\hat{H} = \frac{1}{2m} \left\{ (\vec{p} - q\vec{A})^2 \sigma_0 - i\hbar q \vec{\sigma} \cdot \vec{B} \right\}$$

which is the same as above.

↑  
Spin precession  
term